

GR problem set 03 (Fri. 31.10.2025)

16. **Geodesic equation.** Please derive geodesics equation. **Hint:** consult [geodezyjna.pdf](#) file (comments are in Polish but they should be more or less self explanatory) or go to F. Schuller's lectures.
17. Show the following symmetry properties of the Riemann tensor build from Levi-Civita connection:

- (a) $R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta} = -R_{\alpha\beta\nu\mu} = -R_{\beta\alpha\mu\nu}$,
- (b) $R^\alpha_{\beta\mu\nu} + R^\alpha_{\nu\beta\mu} + R^\alpha_{\mu\nu\beta} = 0$,
- (c) $R^\alpha_{\beta\mu\nu;\lambda} + R^\alpha_{\beta\lambda\mu;\nu} + R^\alpha_{\beta\nu\lambda;\mu} \equiv \nabla_\lambda R^\alpha_{\beta\mu\nu} + \nabla_\nu R^\alpha_{\beta\lambda\mu} + \nabla_\mu R^\alpha_{\beta\nu\lambda} = 0$ (Bianchi identities),
- (d) $(R^{\alpha\beta} - \frac{1}{2}g^{\alpha\beta}R)_{;\beta} = 0$ (contracted Bianchi identities).

Hint: the symmetries (a)-(c) can be easily proven in locally flat coordinates and (d) follows from (c).

Think/check how these properties would be modified in presence of torsion/non metric compatible connection.

18. Please show that in n -dimensional space and in the case of Levi-Civita connection (such that all symmetries from previous exercise hold) the Riemann tensor has $n^2(n^2 - 1)/12$ independent components.
19. Please find all non-vanishing components of Christoffel symbols, Riemann tensor $R_{\alpha\beta\mu\nu}$, Ricci tensor and Ricci scalar for a unit sphere,

$$d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2 = \frac{du^2}{1-u^2} + (1-u^2) d\phi^2, \quad \text{where } u = \cos\theta$$

Hint: You can find Euler-Lagrange equations for the Lagrangian $L = (1/2)(g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta)$ (where $\dot{x}^\mu = dx^\mu/d\lambda$ and λ is an arbitrary parameter) that read:

$$\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$$

and thus read out all Christoffel symbols. In "pen and paper" computations this is usually much more efficient than using the definition for the Christoffel symbols.

20. Of course, having a metric, finding Christoffel symbols, Riemann tensor, etc. is a purely algorithmic task. Get *Mathematica* running on your computers (there should be a student licence available for all of you here).

Try to write a simple *Mathematica* code that does this job, and in particular can check if a given metric solves Einstein equations.

Hint: you can base your method on appendix C in [J.B. Hartle *Gravitation*]. An alternative is to use some packages - google for **grquick** and **diffgeo** packages.

21. Calculate what is the result of a parallel transport of a vector along a parallel at θ_0 for a round sphere ($\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta$, $\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta$ and other Γ 's equal zero).
22. **Parallel transport and curvature.** Show that the parallel transport of a vector Z along infinitesimal parallelogram defined by two vector fields A and B , such that $[A, B] = 0$ (why is this condition necessary?) results in:

$$\delta Z^\mu|_p = R^\mu_{\nu\beta\alpha} Z^\nu B^\beta A^\alpha|_p \lambda_A \lambda_B + \mathcal{O}(\lambda^3) ,$$

where the parallelogram $pqrs$ is such that starting from the point p we can reach the point r either

- (a) by going from p to q moving by λ_A along the vector field A and then from q to r moving by λ_B along the vector field B , or
- (b) by going from p to s moving by λ_B along the vector field B and then from s to r moving by λ_A along the vector field A